

250 CORE JAVA INTERVIEW QUESTIONS

PART 1: Java Basics & Fundamentals

1. What is Java?
2. Why is Java called platform independent?
3. What is bytecode?
4. What is JVM?
5. What is JRE?
6. What is JDK?
7. Difference between JVM, JRE, and JDK?
8. What is JIT compiler?
9. What are the features of Java?
10. What is the main() method?
11. Why is main() static?
12. Can a Java program run without main()?
13. What are identifiers in Java?
14. What are keywords in Java?
15. Is Java 100% object-oriented?
16. What are primitive data types?
17. What is a literal?
18. What is a variable?
19. Local vs instance vs static variables?
20. What is type casting?
21. What is widening and narrowing?

22. What is a command-line argument?
 23. What is a package?
 24. What is the default package?
 25. What is a ClassLoader?
 26. Types of ClassLoaders?
 27. What is a token in Java?
 28. What is a Unicode system?
 29. What is the role of the compiler in Java?
 30. What is a source file?
 31. Why is Java not using pointers?
 32. What is a comment? Types?
 33. What is a constant?
 34. Difference between == and = ?
 35. Can we have multiple main methods?
 36. What is System.out.println()?
 37. Difference between print() and println()?
 38. What is an entry point of Java?
 39. What is a Java IDE?
 40. What is the difference between JDK and JVM?
-

PART 2: OOP Concepts

41. What is OOP?
42. What are the four pillars of OOP?
43. What is a class?

44. What is an object?
45. What is encapsulation?
46. What is abstraction?
47. What is inheritance?
48. What is polymorphism?
49. Types of inheritance?
50. Why doesn't Java support multiple inheritance (classes)?
51. What is IS-A relationship?
52. What is HAS-A relationship?
53. What is method overloading?
54. What is method overriding?
55. Difference between overloading and overriding?
56. Can we overload main()?
57. Can we override static methods?
58. Can we override private methods?
59. What is dynamic method dispatch?
60. What is constructor?
61. Types of constructors?
62. What is constructor overloading?
63. Can constructor be inherited?
64. Can constructor be final?
65. What is this keyword?
66. What is super keyword?
67. Difference between this() and super()?
68. What is a final class?

69. What is a final variable?
 70. What is a final method?
 71. What is an interface?
 72. What is an abstract class?
 73. Difference between abstract class and interface?
 74. Can an interface have variables?
 75. Can an interface have constructors?
 76. What are default methods (in interfaces)?
 77. What are static methods in interfaces?
 78. What is a marker interface?
 79. What is multiple inheritance using interfaces?
 80. What is object class in Java?
-

PART 3: Memory, Variables & Keywords

81. What is heap memory?
82. What is stack memory?
83. Difference between heap and stack?
84. What is a reference variable?
85. When are objects created in memory?
86. What is garbage collection?
87. Can we force garbage collection?
88. What is finalize()?
89. What is static keyword?
90. What is a static block?

91. What is a static method?
92. What is a static variable?
93. Can static methods access non-static variables?
94. What is the purpose of the volatile keyword?
95. What is the transient keyword?
96. What is a mutable object?
97. What is an immutable object?
98. What is shallow copy?
99. What is deep copy?
100. What is cloning?
101. What is the Cloneable interface?
102. What is serialVersionUID?
103. What is a constant pool?
104. What is a symbol table?
105. What is method signature?
106. What is a nested class?
107. What is an inner class?
108. What is a static nested class?
109. What is a local inner class?
110. What is an anonymous inner class?

PART 4: String & Wrapper Classes

111. What is String?
112. Why is String immutable?

113. What is String Constant Pool?
 114. Difference between String and StringBuffer?
 115. Difference between StringBuffer and StringBuilder?
 116. Which is thread-safe: StringBuffer or StringBuilder?
 117. What is intern()?
 118. What is substring()?
 119. How many objects are created in new String("abc")?
 120. Can we extend String class?
 121. What is toString() method?
 122. What is String concatenation?
 123. What is equals() in String?
 124. Difference between equals() and ==?
 125. What is compareTo()?
 126. What are wrapper classes?
 127. What is autoboxing?
 128. What is unboxing?
 129. What is valueOf()?
 130. What is parseInt()?
 131. Can wrapper classes be mutable?
 132. Can we create custom wrapper classes?
 133. What is Character class?
 134. What is Boolean wrapper?
 135. What is Number class?
-

PART 5: Arrays & Collections

136. What is an array?
137. Types of arrays?
138. Default values in arrays?
139. Can arrays be resized?
140. Difference between array and ArrayList?
141. What is Collection framework?
142. What is Collection interface?
143. What is List?
144. What is Set?
145. What is Map?
146. Difference between List and Set?
147. Difference between Set and Map?
148. What is ArrayList?
149. What is LinkedList?
150. ArrayList vs LinkedList?
151. What is Vector?
152. What is Stack?
153. What is Queue?
154. What is Deque?
155. What is PriorityQueue?
156. What is HashSet?
157. What is LinkedHashSet?
158. What is TreeSet?

159. Difference between HashSet and TreeSet?
160. What is HashMap?
161. What is Hashtable?
162. Difference between HashMap and Hashtable?
163. What is LinkedHashMap?
164. What is TreeMap?
165. Difference between HashMap and TreeMap?
166. How does HashMap work internally?
167. What is hashing?
168. What is load factor?
169. What is initial capacity?
170. What is collision?
171. What is chaining?
172. What is Comparable?
173. What is Comparator?
174. Comparable vs Comparator?
175. What is Iterator?
176. What is ListIterator?
177. Iterator vs ListIterator?
178. What is fail-fast iterator?
179. What is fail-safe iterator?
180. What is ConcurrentHashMap?
181. What is CopyOnWriteArrayList?
182. What is Enumeration?
183. Difference between Enumeration and Iterator?

- 184. What is Collections utility class?
 - 185. What is Arrays utility class?
-

PART 6: Exception Handling

- 186. What is an exception?
- 187. What is error?
- 188. Difference between error and exception?
- 189. What is exception hierarchy?
- 190. What are checked exceptions?
- 191. What are unchecked exceptions?
- 192. What is try block?
- 193. What is catch block?
- 194. What is finally block?
- 195. Can finally be skipped?
- 196. What is throw?
- 197. What is throws?
- 198. throw vs throws?
- 199. What is custom exception?
- 200. What is exception propagation?
- 201. Can a constructor throw exception?
- 202. Can we have multiple catch blocks?
- 203. Order of catch blocks?
- 204. What is multi-catch?
- 205. What is try-with-resources?

- 206. What is StackTrace?
 - 207. What is NullPointerException?
 - 208. What is ArrayIndexOutOfBoundsException?
 - 209. What is ClassCastException?
 - 210. What is NumberFormatException?
 - 211. What is IllegalArgumentException?
 - 212. What is IllegalStateException?
 - 213. What happens if exception occurs in finally?
 - 214. Can we catch Throwable?
 - 215. Can we rethrow exception?
-

PART 7: Multithreading & Concurrency

- 216. What is a process?
- 217. What is a thread?
- 218. Difference between process and thread?
- 219. What is multitasking?
- 220. What is multithreading?
- 221. What is concurrency?
- 222. How to create a thread?
- 223. Runnable vs Thread?
- 224. What is start()?
- 225. Can we call start() twice?
- 226. run() vs start()?
- 227. What is sleep()?
- 228. What is join()?

- 229. What is yield()?
 - 230. Difference between wait() and sleep()?
 - 231. Why wait/notify are in Object class?
 - 232. What is synchronization?
 - 233. What is synchronized method?
 - 234. What is synchronized block?
 - 235. What is deadlock?
 - 236. What is starvation?
 - 237. What is livelock?
 - 238. What is daemon thread?
 - 239. What is ThreadLocal?
 - 240. What is thread pool?
-

PART 8: Java 8 Core Features

- 241. What is Java 8?
- 242. What is lambda expression?
- 243. What is functional interface?
- 244. What is @FunctionalInterface?
- 245. What is method reference?
- 246. What is Stream API?
- 247. What is Optional class?
- 248. Difference between map() and flatMap()?
- 249. What is default method in interface?
- 250. What is static method in interface?

-: ALL THE BEST :-